

Machine Learning – Assignment

This is a group assignment.

You may apply a Machine learning algorithm (under supervised, unsupervised or reinforcement learning) to solve any real-world problem based on any real-world dataset. **You should NOT use Deep learning models.** You may use a publicly available dataset from a source like the following one (you can choose any other data source. Be sure to cite where you got your data from. **The dataset has to be real-world data that is published on the internet**). Following are two example datasources for ML datasets. You may use any additional data source as long as they are credible and publicly available.

<http://archive.ics.uci.edu/ml/index.php>

<https://www.kaggle.com/datasets>

Based on a dataset, you may apply machine learning algorithms and then get the test results. You may use Python as the programming language and Jupyter Notebook as the programming environment.

For supervised learning and unsupervised learning, **four** distinct algorithms should be applied on the dataset (team size 4).

For reinforcement learning, **only two** distinct algorithms should be applied (for teams with three or four members both).

Submission

- The ids, emails of the members in a text file called 'members.txt'.
- A text file called 'submission.txt' containing the link to the dataset. If you're using your own dataset, make sure to upload to a shareable cloud directory and copy the link here.
- Link to a public Github repository containing the Jupyter Notebooks containing the code appended to the same 'submission.txt' file. **The github repository should contain the commit history with detailed commits to be considered as a valid submission.**
- Link to a youtube video (maximum video length is 20 minutes) containing the presentation/demo of the work done. Each member can use maximum 4 mins to explain their contribution to the project, appended to the same 'submission.txt' file.
- A report in **pdf format** with a description on the problem addressed, dataset used, methodology (algorithms used), results and discussion (including possible limitations/future work). **The report should compare and contrast the different algorithms used, in terms of their accuracy, using appropriate metrics.**
- The source code should be added as an appendix to the report.
- The entire submission bundle has to be uploaded to the courseweb link (which will be available close to the deadline) as a single zip file. The name of the zip file should be ML-assignment. **Only one person needs to upload the submission on behalf of the group.**

The report should have an appendix that contains all the source code (**added as text, not screen-shots**).
If the source code is not added as text in the appendix, it won't be accepted as a valid submission.

Both the report and the source code should be there to be accepted as a valid submission.

The github repository should contain the commit history with detailed commits to be considered as a valid submission.

Please note if any of the above components are missing in the submission, it will not be accepted as a valid submission and will be given zero marks. Any resubmissions will be awarded a maximum mark of 45%.

All reports will be uploaded to Turnitin for plagiarism checking. **If the turnitin similarity is above 20%, marks will be penalized.**

Deadline: End of 10th week of the semester

Marks allocated: 20